

QA Report: Collections API

Report ID: QA-PEG-005 | **Date:** 2026-04-09 | **Sprint:** Sprint 3 **Service:** Ubn-Collections-API | **Environment:** Development (dev-api.onecluster.co)

Executive Summary

The Collections API provides static virtual collection account creation, inbound transaction history retrieval, and webhook registration for real-time payment notifications. In sandbox mode, the service returns SB-prefixed account numbers and deterministic mock data. This report covers QA testing of all Collections API endpoints on the development environment.

Scope

- POST /api/v1/collections/accounts — Create static virtual collection account
- GET /api/v1/collections/accounts/{accountNumber}/transactions — Inbound transaction history
- POST /api/v1/collections/webhooks — Register webhook endpoint (with ping validation)
- Sandbox behavior: SB-prefixed account numbers, mock transactions
- HMAC-SHA256 signature validation
- Webhook ping validation and retry logic
- Error handling and field naming consistency

Changes This Sprint (Mar 31 — Apr 9)

- 17 commits to Ubn-Collections-API
- CI/CD pipeline configuration and automated deployment
- VPS deployment setup
- Webhook ping validation implemented (POST to partner URL with challenge)
- Sandbox mock response improvements
- Response envelope standardization
- Inbound transaction event logging
- Health check and readiness probe endpoints added

Test Results

Virtual Collection Account Creation

Test ID	Test Case	Expected	Actual	Status	Notes
QA-COL-001	Create collection account with valid payload	201 with accountNumber	201 with virtualAccountNumber (SB-prefixed), accountName, bankName	PASS	Sandbox returns SB- prefix for easy identification
QA-COL-002	Create collection account with missing fields	422 Validation error	422 with field-level errors (accountName, customerEmail required)	PASS	Proper RFC 7807 format
QA-COL-003	Create collection account without HMAC	401 HMAC_REQUIRED	401 with error:"HMAC signature is required"	PASS	Security gate enforced
QA-COL-004	Create collection account — field name check	accountNumber in response	virtualAccountNumber returned instead of accountNumber	FAIL	See ISSUE PEG-383 — field name mismatch with spec

QA-COL-005	Create duplicate collection account (same reference)	409 or idempotent 200	200 with existing account returned	PASS	Idempotency on reference key
QA-COL-006	SB- prefix on sandbox accounts	Account number starts with SB-	"SB-1234567890" format confirmed	PASS	Clear sandbox identification

Inbound Transaction History

Test ID	Test Case	Expected	Actual	Status	Notes
QA-COL-007	Get transactions for valid account	200 with transactions array	200 with transactions array, pagination metadata	PASS	Returns seeded sandbox transactions
QA-COL-008	Get transactions for non-existent account	404 ACCOUNT_NOT_FOUND	404 with error:"Account not found"	PASS	Proper error handling
QA-COL-009	Get transactions with date range	200 with filtered results	200 with transactions filtered by startDate/endDate	PASS	Date filtering works
QA-COL-010	Get transactions with pagination	200 with page metadata	200 with page, pageSize, totalCount, totalPages	PASS	Pagination metadata present
QA-COL-011	Transaction response — double nesting	data.transactions	data.data.transactions (double-nested)	FAIL	See ISSUE PEG-389 — response wrapped twice
QA-COL-012	Transaction fields present	amount, reference, narration, timestamp	All fields present in each transaction object	PASS	Complete transaction data

Webhook Registration

Test ID	Test Case	Expected	Actual	Status	Notes
QA-COL-013	Register webhook with valid URL	200 with webhookId	200 with webhookId, status:"ACTIVE"	PASS	Webhook registered and ping sent
QA-COL-014	Register webhook — ping validation	Platform sends POST to URL with challenge	Platform sends {event:"ping", challenge:""}, expects challenge echoed back	PASS	Ensures partner URL is live and responsive
QA-COL-015	Register webhook with unreachable URL	422 WEBHOOK_VALIDATION_FAILED	422 with error:"Webhook URL did not respond to ping"	PASS	Dead URLs rejected at registration

QA-COL-016	Register webhook with non-HTTPS URL	422 HTTPS_REQUIRED	422 with error:"Webhook URL must use HTTPS"	PASS	Security requirement enforced
QA-COL-017	Register webhook without auth	401 Unauthorized	401 with error:"API key is required"	PASS	Auth enforcement verified
QA-COL-018	Duplicate webhook registration	409 or update existing	200 with existing webhook updated	PASS	Upsert behavior on same URL

Webhook Delivery (Code Review)

Test ID	Test Case	Expected	Actual	Status	Notes
QA-COL-019	Webhook retry on failure	Exponential backoff, max 5 retries	Code review: retry policy with 30s, 60s, 120s, 300s, 600s intervals	CODE REVIEW PASS	Dead letter queue after 5 failures
QA-COL-020	Webhook payload signature	HMAC-SHA256 in X-Webhook-Signature header	Code review: webhook payload signed with partner's secret key	CODE REVIEW PASS	Partners can verify authenticity
QA-COL-021	Webhook payload contents	transactionRef, amount, accountNumber, timestamp	Code review: all fields included in webhook payload	CODE REVIEW PASS	

Issues Found

ID	Severity	Issue	Status
PEG-383	Medium	Response field name is virtualAccountNumber but OpenAPI spec defines accountNumber — inconsistency will cause partner integration failures if SDK/docs use spec field names	OPEN
PEG-389	Medium	Inbound transaction history response has double nesting: data.data.transactions instead of data.transactions — response envelope is wrapped twice	OPEN
PEG-398	Low	Webhook ping timeout is 5 seconds — may be too aggressive for partners with cold-start serverless functions	OPEN

Field Name Discrepancy Detail (PEG-383)

OpenAPI Spec:

```
{
  "data": {
    "accountNumber": "SB-1234567890",
    "accountName": "Test Collection Account",
    "bankName": "Union Bank"
  }
}
```

Actual Response:

```
{
  "data": {
    "virtualAccountNumber": "SB-1234567890",
    "accountName": "Test Collection Account",
    "bankName": "Union Bank"
  }
}
```

Partners building against the published spec will reference `data.accountNumber` and receive `undefined`. Either the spec or the API must be updated to match.

Double Nesting Detail (PEG-389)

Expected Response:

```
{
  "isSuccess": true,
  "data": {
    "transactions": [...],
    "page": 1,
    "pageSize": 20,
    "totalCount": 5
  }
}
```

Actual Response:

```
{
  "isSuccess": true,
  "data": {
    "data": {
      "transactions": [...],
      "page": 1,
      "pageSize": 20,
      "totalCount": 5
    }
  }
}
```

The response envelope wrapper is applied twice — likely the controller returns a `ServiceResponse<PagedResult>` which is then wrapped again by a global response filter.

Recommendations

1. **Fix PEG-383:** Align field name to either `virtualAccountNumber` or `accountNumber` consistently across spec and implementation before partner onboarding
2. **Fix PEG-389:** Remove double nesting in transaction history response — likely a middleware/filter issue wrapping the response envelope twice
3. Increase webhook ping timeout from 5s to 10s to accommodate serverless cold starts
4. Add webhook event type filtering so partners can subscribe to specific event types (e.g., only CREDIT events)
5. Document SB- prefix convention in partner integration guide
6. Consider adding a webhook test/replay endpoint for partners to re-trigger missed notifications during development